

Vertical Niches Partitioning of Soil Fungal Communities in Esker Forests

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INTRODUCTION

Eskers are **fluvioglacial geological formations** providing many **natural resources** important for the economy and the population.

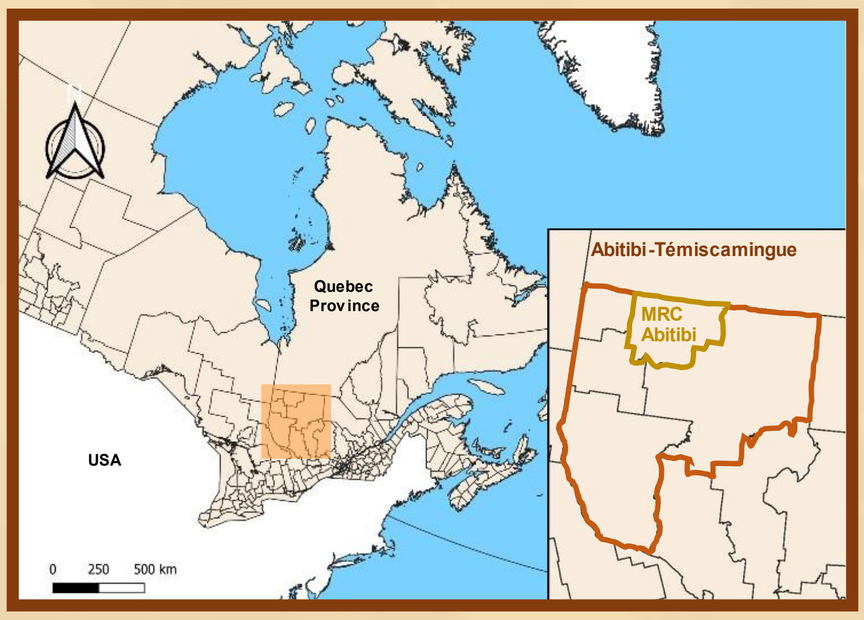
The associated fungal communities have barely been investigated to date, notably **vertical spatial patterns.**

Esker forest ecosystems are **vulnerable** as they are highly sought over for water, wood, sand and gravel.

Need to **fill the gap** in our fundamental knowledge and understanding of these communities.

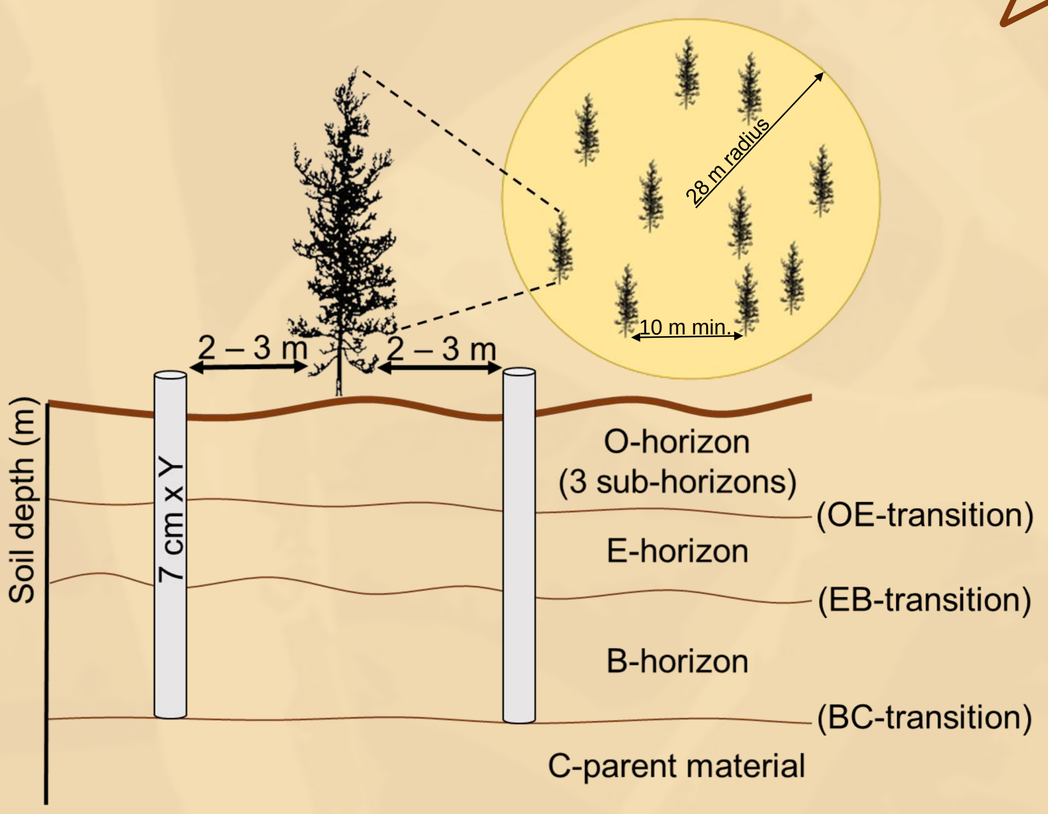
MATERIAL AND METHODS

Study area



Esker sites will be selected in the **Abitibi region** characterized by **numerous** totally or partially **emerged esker formations** within the glaciolacustrine matrix.

Soil sampling

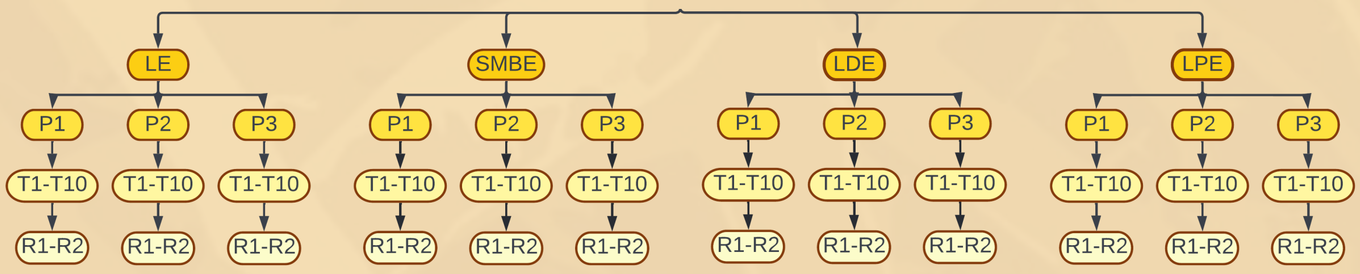


Total of **240 soil cores,** resulting in **72 pooled soil samples.**

OBJECTIVES

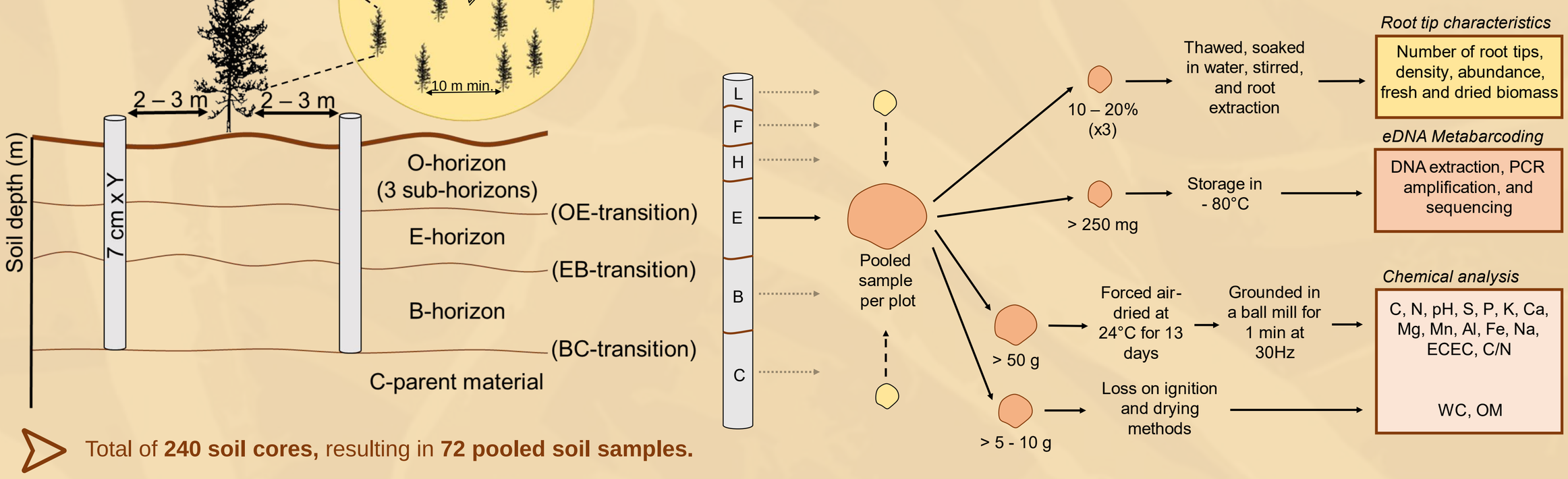
- Characterize the **diversity**, **composition**, and **structure** of **esker podzol fungal communities** across **soil horizons.**
- Determine the **key abiotic** and **biotic factors** that shape these **belowground communities** on a **vertical axis.**

Experimental design: Esker*Funga*



Composed of **12 (4 eskers x 3 plots) circular plots** (2,500 m² each, total of 30,000 m²) with a radius of 28 m containing **10 randomly selected trees** (total of 120) at least 10 m from each other with **2 soil core replicates per tree**.

Data analyses



ORIGINALITY

- First generated, **molecular-based** fungal data with such **inclusive vertical sampling** for this kind of ecosystem worldwide.

CONTRIBUTIONS

- Understand the **diversity, vertical distribution** and **driving factors** of **soil fungal communities** in **esker forests.**

Personal Website

ResearchGate